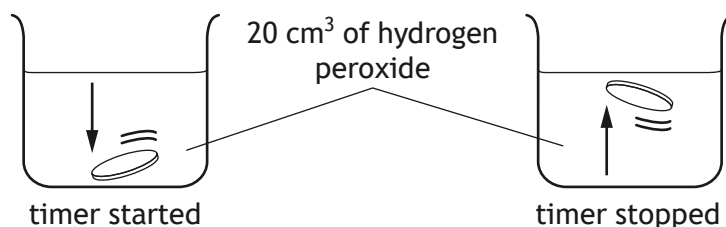


5. An investigation was carried out into the effect of substrate concentration on the inhibition of catalase by copper nitrate. Catalase breaks down hydrogen peroxide into water and oxygen.

10 cm³ of catalase solution was mixed with 10 cm³ copper nitrate solution and a paper disc was placed in this solution for 10 seconds.

The disc was then dropped into hydrogen peroxide. When the disc reached the bottom of the beaker, a timer was started and the time taken for the disc to float to the surface was recorded. Oxygen produced during the reaction caused the paper disc to float.



This was repeated using different concentrations of hydrogen peroxide.

A control experiment was carried out where copper nitrate solution was replaced with water.

The results are shown in the table.

Concentration of hydrogen peroxide (%)	Time for disc to float to surface (seconds)	
	Catalase + copper nitrate solution	Catalase solution
0.5	192	120
1.0	136	84
2.0	72	36
3.0	52	28
5.0	28	28

- (a) Name the type of metabolic reaction catalysed by catalase.

1

- (b) (i) State two variables, not already mentioned, that should be controlled to make the investigation valid.

2

1. _____

2. _____

- (ii) State the purpose of the control in this investigation.

1



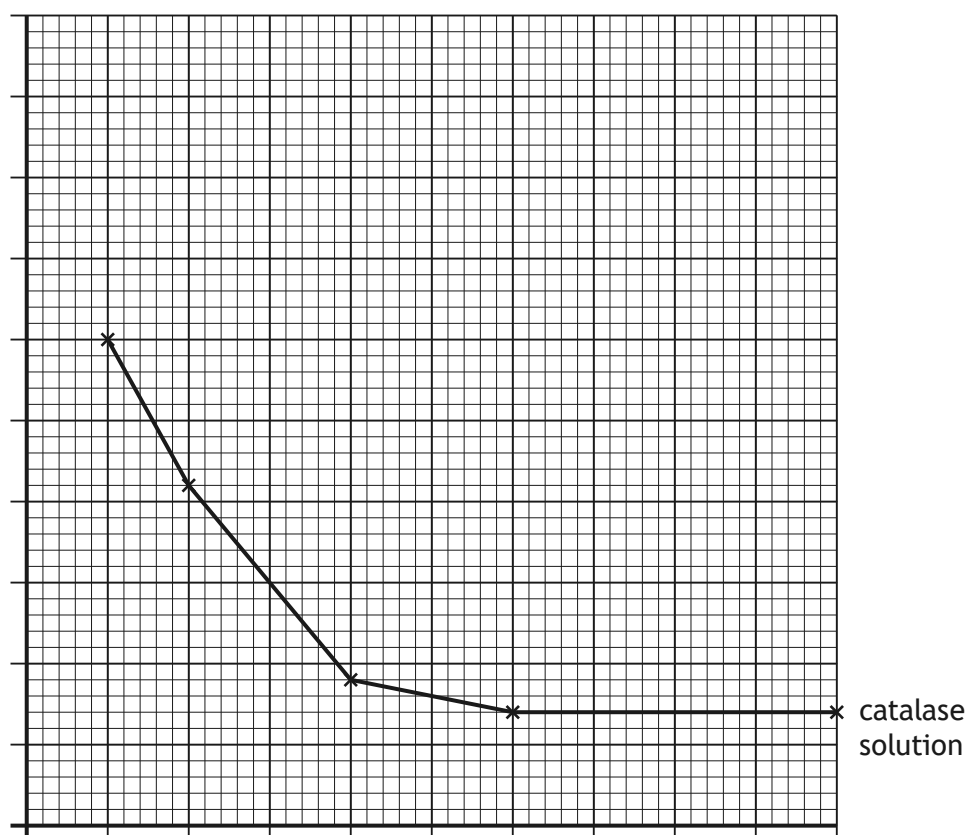
5. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

- (c) On the grid, complete the line graph to show the results of the investigation using **catalase** and **copper nitrate**.

2

(Additional graph paper, if required, can be found on *page 28*.)



- (d) State the conclusion that can be made from these results.

1

- (e) Another disc was placed in the solution of catalase and copper nitrate then dropped into a beaker of 6.0% hydrogen peroxide.

Predict the average time for the disc to float to the surface.

1

_____ seconds

[Turn over

